Case Study: Reducing Food Waste in School Cafeteria

1. Introduction

Food waste is a significant issue in schools, leading to **economic loss, environmental impact, and missed opportunities to help those in need**. In our school cafeteria, observations show that a large portion of meals served daily is discarded, contributing to unnecessary waste. This case study analyzes the problem, explores multiple solutions using creative thinking, evaluates them, and recommends an effective strategy.

2. Problem Analysis

- **Problem Statement:** The school cafeteria generates excessive food waste every day, with an estimated **30% of meals uneaten and discarded**.
- **Impacted Stakeholders:**
  - Students (wasted resources that could be better used)
  - Cafeteria staff (extra work handling waste)
  - School administration (financial losses)
  - Environment (contributes to landfill waste)
- **Root Causes Identified:**
  1. Students overestimating how much food they can eat.
  2. Menu portions are too large or not tailored to preferences.
  3. Lack of awareness about food waste consequences.
  4. Inefficient inventory and meal planning.

3. Possible Solutions (Creative Thinking)

1. **Adjust Portion Sizes** – Offer smaller portion options and allow students to choose additional servings if needed.
2. **Implement "Take-What-You-Can-Eat" Policy** – Encourage students to serve themselves based on appetite.
3. **Awareness Campaigns** – Educate students about environmental and financial impact of food waste through posters, workshops, and competitions.
4. **Food Donation Program** – Collect leftover, safe-to-eat food and donate it to local shelters.
5. **Digital Feedback System** – Students rate menu items daily to adjust future meal planning according to preferences.

4. Evaluation of Solutions

Solution	Pros	Cons	Feasibility	Impact
Adjust Portion Sizes	Reduces waste immediately; flexible	Requires staff training	High	Medium
"Take-What-You-Can-Eat" Policy	Empowers students; reduces leftovers	May slow serving time	Medium	High
Awareness Campaigns	Long-term behavioral change	Slow results; needs effort	High	Medium
Food Donation Program	Reduces waste; helps community	Logistics needed; liability issues	Medium	Medium
Digital Feedback System	Data-driven menu planning	Needs tech setup	Medium	High

Analysis:

- Combining **adjusted portion sizes**, **take-what-you-can-eat policy**, and **awareness campaigns** is the most feasible and high-impact approach.
- Donation and digital feedback can be added gradually for long-term improvement.

5. Recommendation

To effectively reduce food waste, the school should implement a **combined strategy**:

1. **Smaller portion options** with refill possibility.
2. **Self-service policy** to match individual appetite.
3. **Student awareness campaigns** highlighting environmental and financial impacts.

This strategy is **cost-effective, practical, and directly targets the root causes** while encouraging responsible behavior.

6. Conclusion

Through systematic problem analysis and creative solution generation, the school can significantly **reduce food waste**, save costs, and educate students about sustainability. This case study demonstrates that breaking down the problem, brainstorming multiple solutions, and evaluating them against clear criteria leads to **effective and sustainable problem-solving**.